

# LEARNED NYQUIST: FUNDAMENTAL ALIASING LIMITS OF IMPLICIT NEURAL REPRESENTATIONS

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## ABSTRACT

Implicit neural representations (INRs)—SIREN and Fourier-feature coordinate networks—are increasingly used as continuous models of signals. A recent *achievability* sampling theory addresses signals that are themselves INRs; we address the *converse* question that governs practice, where the signal is arbitrary and observed at finite, noisy, nonuniform samples. A fixed-feature INR (fixed feature frequencies, linear output layer) represents only a *structured* frequency set  $\Lambda$ . Casting its least-squares fit as generalized sampling on this dictionary, we prove: (i) achievability—if the signal spectrum lies in  $\Lambda$  and the samples render the dictionary matrix full column rank, a bandwidth-matched INR recovers it with closed-form noise gain  $\kappa = 1/\sigma_{\min}(\Phi)$ ; (ii) a converse—out-of-band energy is *folded* onto  $\Lambda$  with an error floor no fixed-feature INR can beat, and the sample residual can be far smaller than the resulting spectral corruption (*silent aliasing*), with a *non-classical* fold target when  $\Lambda$  is structured; (iii) an average-case decomposition of the estimator’s error into an estimation-variance term and an aliasing-variance floor that persists under coherent (commensurate) folds. We give a bandwidth-matched estimator and a ground-truth-free diagnostic, validate every identity against Monte-Carlo, and demonstrate both sides of the limit on real audio and scientific signals—the phenomenon persisting for trained *nonlinear* and 2-D INRs. The classical uniform-lowpass fold is one special case; the structured/learned aliasing, the named silent-aliasing regime, and the diagnostic are new.

**Index Terms**— Implicit neural representations, sampling theory, aliasing, generalized sampling, coordinate networks.

## 1. INTRODUCTION

Implicit neural representations parameterize a signal as a coordinate network  $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$  and are now a default continuous model for audio, images, volumes, and scientific fields [1, 2, 3]. A large literature engineers their *architectures*—periodic activations, Fourier features, band-limited and anti-aliased variants [1, 2, 4, 5, 6]—and a recent thread gives a *sampling theory*: Najaf and Ongie [7] and the super-resolution follow-up [8] prove that a signal which is a shallow INR is exactly recoverable from enough noiseless Fourier measurements. These are *achievability* statements about realizable signals [9].

*The gap.* In practice the signal is not an INR: it is an arbitrary waveform observed at finitely many, often nonuniform, noisy samples. Nothing above says when the task is possible, how the answer depends on the network’s capacity, or what failure looks like. We supply the converse. The key object is the INR’s *representable frequency set*  $\Lambda$ : for a Fourier-feature/periodic coordinate network the

output is a combination of complex exponentials whose frequencies are integer combinations of the base feature frequencies—a set that grows with width and depth [10]. A signal is representable only insofar as its spectrum lies in  $\Lambda$ .

*Scope and contributions.* Our theorems concern the analytically exact case: a fixed-feature INR whose output layer is linear over  $\Lambda$ . Its least-squares fit is generalized (nonharmonic) sampling on the finite dictionary  $\Lambda$  [11, 12]; that recast is the mechanism. On top of it we establish the INR-specific facts: (1) an achievability bound with a closed-form noise gain (Thm. 1); (2) a converse in which out-of-band energy folds onto  $\Lambda$  with an error floor, producing *silent aliasing*—small sample residual, large spectral corruption—and, for structured  $\Lambda$ , a fold target unlike the classical lowpass law (Thm. 2); (3) an average-case error decomposition separating estimation variance from an aliasing-variance floor that persists under coherent folds (Thm. 3); and (4) a bandwidth-matched estimator, a ground-truth-free diagnostic, Monte-Carlo-validated theory, and real-data demonstrations of both sides, which persist empirically for trained nonlinear and 2-D INRs.

*Relation to Nyquist.* When  $\Lambda = \{-B, \dots, B\}$  and sampling is uniform, our converse reduces to classical aliasing (Rem. 1). The novelty is that  $\Lambda$  is a *structured* (and, for adaptive-frequency INRs, signal-dependent) set, so energy folds onto a structured target; the named silent-aliasing regime, the data-only diagnostic, and the matched estimator have no lowpass analogue.

## 2. SIGNAL MODEL AND REPRESENTABLE BAND

Let  $f : [0, 1) \rightarrow \mathbb{R}$  be observed at samples  $t_1, \dots, t_N$  (possibly nonuniform) as  $y_j = f(t_j) + \varepsilon_j$ . A fixed-feature INR with feature set inducing  $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$  has model class  $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$  where  $e_\omega(x) = e^{i2\pi\omega x}$ ; deeper/periodic networks enlarge  $\Lambda$  to integer combinations of base frequencies [10]. Define the *synthesis matrix*  $\Phi \in \mathbb{C}^{N \times m}$ ,  $\Phi_{jk} = e_{\omega_k}(t_j)$ ; fitting the INR is the least-squares problem  $\hat{c} = \arg \min_c \|\mathbf{y} - \Phi \mathbf{c}\|^2$  (the code adds a negligible ridge for numerics). Split  $f = f_{\text{in}} + f_{\text{out}}$  with  $\text{supp}(\mathcal{F}f_{\text{in}}) \subseteq \Lambda$  ( $\mathcal{F}$  the Fourier transform) and  $f_{\text{out}}$  spectrally disjoint from  $\Lambda$  (frequencies  $\nu_l$ , synthesis matrix  $\Phi_{\text{out}}$ ), so  $\mathbf{y} = \Phi \mathbf{c}^* + \mathbf{g} + \varepsilon$  with  $g_j = f_{\text{out}}(t_j)$ . The lower *frame bound* is  $A = \sigma_{\min}(\Phi)^2$ ;  $A > 0$  iff  $\Phi$  has full column rank ( $m \leq N$ , generic positions), i.e. the samples determine  $\mathcal{M}_\Lambda$ .

## 3. FUNDAMENTAL LIMITS

**Theorem 1** (Achievability). *If  $f_{\text{out}} = 0$  and  $\Phi$  has full column rank ( $\sigma_{\min}(\Phi) > 0$ ), then  $\|\hat{c} - \mathbf{c}^*\| \leq \kappa \|\varepsilon\|$  with noise gain  $\kappa = \|(\Phi^* \Phi)^{-1} \Phi^*\|_2 = 1/\sigma_{\min}(\Phi)$ , and for white noise of variance  $\sigma^2$ ,  $\mathbb{E}\|\hat{c} - \mathbf{c}^*\|^2 = \sigma^2 \text{tr}((\Phi^* \Phi)^{-1})$ .*

Full column rank needs  $N \geq m = |\Lambda|$ ; for the special case  $\Lambda = \{-B, \dots, B\}$  this coincides with the Landau density  $N \gtrsim 2B$  [13], but for a sparse structured  $\Lambda$  the binding requirement is  $N \geq |\Lambda|$ , not the band extent.

**Theorem 2** (Converse and aliasing). *Let  $P = \Phi(\Phi^* \Phi)^{-1} \Phi^*$ . With  $f_{\text{out}} \neq 0$  the estimate is biased,  $\hat{c} = c^* + \Delta c$ ,  $\Delta c = (\Phi^* \Phi)^{-1} \Phi^* g$ , and: (i) no INR in  $\mathcal{M}_\Lambda$  fits the noiseless samples better than  $\min_c \|y - \Phi c\| = \|(I - P)g\|$ ; (ii) the recovered spectrum is corrupted by  $\Delta c$ , and as an out-of-band tone a  $e_\nu$  of amplitude  $a$  approaches an atom ( $\nu \rightarrow \omega_k$ ) the residual  $\|(I - P)g\| \rightarrow 0$  while  $\|\Delta c\| \rightarrow |a|$  (silent aliasing: the fit understates the corruption); (iii) the fold target is  $\omega_{k^*}$ ,  $k^* = \arg \max_k \|\Delta c(\nu)\|_k$ .*

**Remark 1.** For uniform  $t_j = j/N$  and  $\Lambda$  an integer lowpass band, (iii) reduces to  $\nu \bmod N$ : classical aliasing is the coherent special case. For structured  $\Lambda$  and nonuniform samples the fold target is non-classical (Fig. 1).

**Theorem 3** (Average-case decomposition). *Under a zero-mean prior on  $c^*$ , out-of-band coefficients of power  $p$ , and white noise  $\sigma^2$ , the mean-squared error of the least-squares estimator is*

$$\mathbb{E} \|\hat{c} - c^*\|^2 = \underbrace{\sigma^2 \text{tr}((\Phi^* \Phi)^{-1})}_{\text{estimation variance}} + \underbrace{\text{tr}(M \Sigma_g M^*)}_{\text{aliasing variance}}, \quad (1)$$

with  $M = (\Phi^* \Phi)^{-1} \Phi^*$  and  $\Sigma_g = \Phi_{\text{out}} \text{diag}(p) \Phi_{\text{out}}^*$ . The first term vanishes as the sampling density grows; the second persists when the out-of-band content is commensurate with the sampling (coherent fold) and otherwise decays as  $O(1/N^2)$ . Their crossover  $N^*$  equates the two terms of (1).

The proofs are elementary least-squares/projection identities; each is asserted against Monte-Carlo in the released test suite.

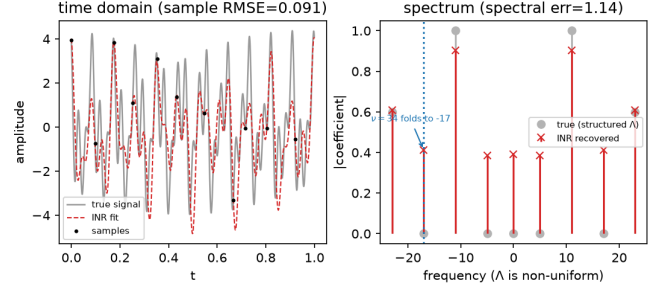
#### 4. MATCHED ESTIMATOR AND DIAGNOSTIC

**Bandwidth matching.** Estimating the essential band from the samples via a nonuniform periodogram and choosing  $\Lambda$  to cover it places the signal in the Thm. 1 regime; this is the estimator we use. **Diagnostic.** Because a coherent fold is nearly invisible in-sample (Thm. 2(ii)), we test band-limitation to  $\Lambda$  by fitting an enlarged dictionary  $\Lambda \cup \Lambda_{\text{ring}}$  ( $\Lambda_{\text{ring}}$  a ring of candidate frequencies outside  $\Lambda$ ) and measuring the recovered energy on the ring; significant ring energy flags out-of-band content with no ground truth.

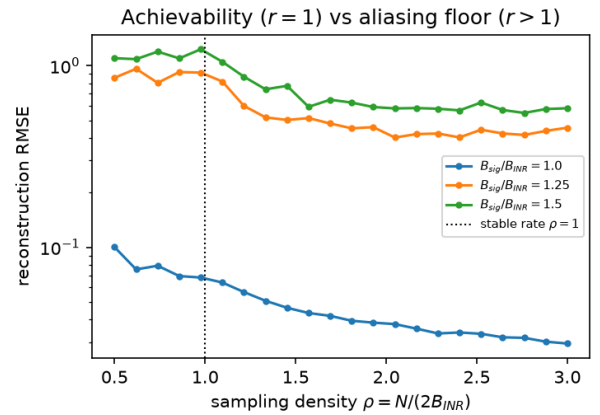
#### 5. EXPERIMENTS

Every identity is validated against Monte-Carlo (frame bounds,  $\kappa$ , aliasing floor, fold law, the Thm. 3 terms). Fig. 1 shows the structured, nonuniform silent-aliasing regime: the fixed-feature INR fits the samples with small residual (RMSE 0.09) yet plants a spurious peak at the non-classical fold frequency; the ring diagnostic of Sec. 4 flags it, placing 58% of the recovered energy outside  $\Lambda$  (released results). Fig. 2 (a 40-trial average) shows the phase transition: an in-band signal's error decays toward the noise floor as sampling density grows (achievability), while any out-of-band content pins the error at a floor independent of density (converse; the density-independent height is the unrepresentable energy  $\|f_{\text{out}}\|$ ).

**Real data, both sides** (Fig. 3). Sweeping the INR bandwidth  $B_{\text{INR}}$  on a real signal and reading the reconstruction-error knee gives the empirical learned-Nyquist rate. Mauna Loa CO<sub>2</sub> (NOAA), after linear detrending, is bandlimited (annual cycle), giving a *sharp*



**Fig. 1.** Silent aliasing onto a structured  $\Lambda = \{0, \pm 5, \pm 11, \pm 17, \pm 23\}$  with nonuniform samples (Thm. 2). The INR fits the samples (RMSE 0.09) but its spectrum is corrupted (error 1.14): the out-of-band tone  $\nu=34$  folds onto the non-classical atom  $-17$ , while the true tones ( $\pm 11, \pm 23$ ) are kept.



**Fig. 2.** Achievability vs. converse (Thm. 1–2), 40-trial Monte-Carlo,  $r = B_{\text{sig}}/B_{\text{INR}}$  the ratio of signal bandwidth to INR representable bandwidth. In-band ( $r=1$ ) error falls toward the noise floor with sampling density; out-of-band ( $r>1$ ) error plateaus at an aliasing floor no density removes.

*knee* at its spectral peak: a narrow INR aliases it, a matched INR recovers it (RMSE 0.41  $\rightarrow$  0.16). Broadband speech (Open Speech Repository) has no clean knee and the error never reaches the noise floor within any practical width—the aliasing side. Smoothed sunspots (SILSO; the smoothing is a documented lowpass proxy) behave like CO<sub>2</sub>. The phenomenon persists for trained *nonlinear* INRs—a shallow Fourier-feature network fits undersampled data but folds an out-of-hull tone in as the linear theory predicts (and a bandwidth-covering network recovers the signal, matching the linear estimator)—and for a 2-D INR on sparse image pixels (Fig. 4): an *over-scaled* representable band silently aliases—it fits the observed pixels to  $\approx 50$  dB (released results) while the full field is moiré-corrupted (16 dB, Fig. 4)—whereas a band matched to the sampling is safe. The single-realization figures are illustrative; the phase transition is the averaged result.

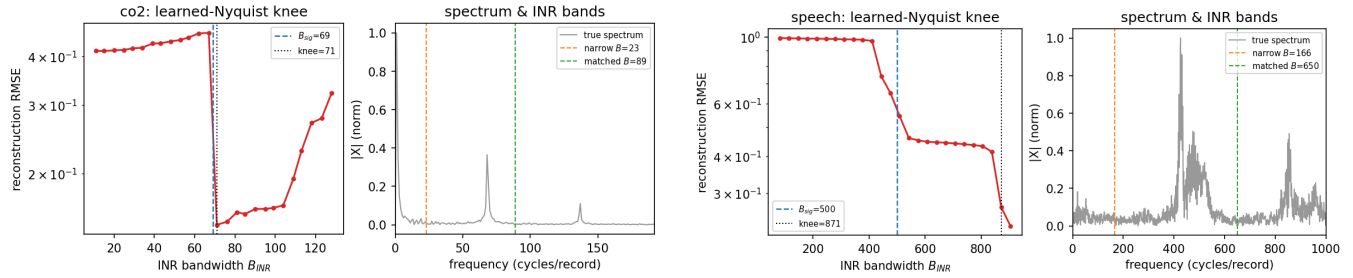
#### 6. CONCLUSION

Fixed-feature INRs alias, and provably so: they represent only a structured band  $\Lambda$ , out-of-band energy folds onto  $\Lambda$  with an error

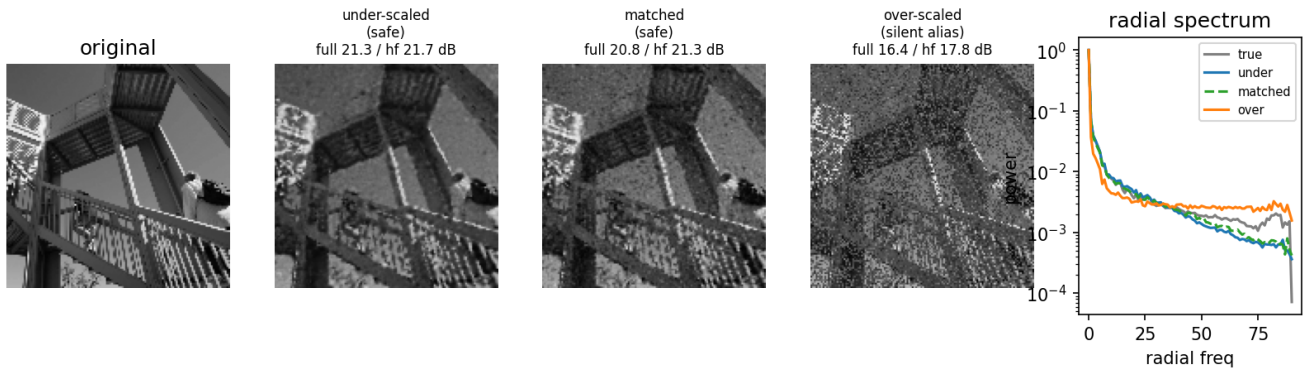
floor, and the corruption can be spectrally severe yet nearly invisible in the sample fit. The limit is sharp, matched by a bandwidth-selecting estimator, detectable from data, and visible on real signals, persisting empirically for trained nonlinear and 2-D INRs. Extending the converse to adaptive-frequency  $\Lambda$  (where the network can move atoms onto the signal) and to the exact training dynamics are the natural next steps.

## 7. REFERENCES

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**Fig. 3.** Both sides of the limit on real data. *Left* ( $\text{CO}_2$ , recoverable): the reconstruction-error knee sits at the annual-cycle frequency; a narrow INR aliases, a matched INR recovers. *Right* (speech, aliasing): a broadband signal has no clean knee and the error never reaches the noise floor within any practical INR width.



**Fig. 4.** The limit generalizes to a trained *nonlinear* 2-D INR fit to sparse image pixels. An over-scaled representable band silently aliases: it fits the observed pixels ( $\approx 50$  dB sample-PSNR; released results) but the full field is moiré-corrupted and its radial spectrum carries spurious high-frequency energy; a band matched to (or below) the sampling is safe.